

SAMA MOHAMED RASHAD EL-HABYAN

Electronics & Communications Engineering Student

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PROFESSIONAL SUMMARY

Electronics & Communications Engineering student at Tanta University with practical experience in embedded systems, microcontroller programming, and engineering simulation. Skilled in Python and Embedded C, with experience using Arduino, ESP32, and AVR microcontrollers. Currently expanding technical knowledge in computer networking and network infrastructure through ongoing training at NTI.

EDUCATION

Tanta University

Bachelor of Engineering in Electronics & Communications Engineering
Third-Year Student

Expected Graduation: 2028

TECHNICAL SKILLS

Programming: Python, Embedded C

Embedded Systems: AVR ATmega32, Arduino, ESP32, Microcontroller Interfacing

Engineering & Simulation: MATLAB, Proteus, NI Multisim

PROJECTS

Smart Load Identification System

Team Project | ESP32, Python, Bluetooth, Feature Extraction, Machine Learning

- Implemented real-time voltage and current data acquisition using ESP32 and electrical sensors.
- Established Bluetooth communication between the ESP32 and computer for real-time data transmission.
- Used Python to process and analyze collected electrical measurements and extract relevant electrical features.
- Applied an existing machine learning classification implementation to the collected dataset to evaluate load identification.

Smart Lock System

Individual Project | AVR ATmega32, Embedded C, Keypad, LCD, EEPROM, Proteus

- Designed and implemented a password-based smart lock system using the AVR ATmega32 microcontroller.
- Programmed the system in Embedded C, implementing keypad, LCD, EEPROM, motor, and buzzer interfacing.
- Implemented password storage and update functionality using EEPROM for data retention.
- Developed security mechanisms including lockout periods and alarm activation after repeated incorrect password attempts.
- Simulated and tested the complete system using Proteus, including the door-opening mechanism.

Sound Frequency Monitoring System

Team Project | Arduino, MATLAB, Signal Processing, FFT Analysis

- Developed a real-time system for monitoring sound frequency and detecting abnormal variations through acoustic signal analysis.
- Integrated a microphone sensor with Arduino for real-time sound signal acquisition.
- Used MATLAB for signal visualization and frequency analysis, including FFT-based analysis and Zero-Crossing frequency estimation.
- Implemented visual and audible alert mechanisms using LED indicators and a buzzer for detected abnormal conditions.

Study Mate

Team Project | Python, Tkinter, File Handling

- Developed the To-Do List module for a student productivity application using Python and Tkinter.
- Implemented task management functionality, including saving and loading tasks using file-based storage.
- Designed and developed the graphical user interface components for the task management section.

TRAINING

Microcontroller-Based Systems Training — NTI / ITIDA

Jun–Jul 2025

Summer Training | 120 Hours | Score: 89%

- Developed practical experience with AVR ATmega32 microcontrollers using Embedded C.
- Worked with microcontroller peripherals and interfacing, including timers, ADC, LCD, and keypad.
- Applied embedded programming concepts through practical implementation and simulation.

Network Infrastructure Training — NTI

Ongoing

Currently developing foundational knowledge in computer networking and network infrastructure.

LANGUAGES

Arabic: Native English: Intermediate